

MH2100 Calculus III

Midterm II (Solutions)

AY 2023/2024, Semester 1

Quantitative Research Society @NTU

October 2023

Overview

- **Coverage:** Lagrange multipliers; double integrals and reversing order; polar coordinates; and volume by a height double integral.
- **Historical source:** MH2100 Midterm II, AY2023/2024 Semester 1. Every question below is transcribed from the official paper. Solutions and marking schemes are unofficial QRS@NTU material.
- **Current administration (AY2026/2027):** Tuesday 6 October 2026, 16:30–18:20 (110 minutes), venue TBA. One double-sided typed A4 sheet is permitted; attachments and sticky notes are not.
- **2026 scope note:** explicit triple integrals, cylindrical and spherical coordinates, general Jacobian transformations, and line integrals are beyond the current midterm scope. They are retained and labelled for historical coverage.

Instructions to candidates (historical, verbatim)

1. DO NOT TURN OVER PAPER UNTIL INSTRUCTED.
2. This midterm paper contains THREE (3) questions.
3. Answer ALL questions. The marks for each question are indicated at the beginning of each question.
4. Candidates should write their answers in the separate answer book provided.
5. ONE two-sided A4 piece of paper with handwritten notes is permitted.

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Question 1**(17 Marks)****QUESTION 1.**

- (i) [5 marks] A wire has a shape of an ellipse described by $4x^2 + y^2 = 8$. The temperature $T(x, y)$ at every point (x, y) of the wire is described by $T(x, y) = e^{2xy}$. Use the method of Lagrange multipliers to find the point(s) on the wire where the wire has the highest and the lowest temperature.
- (ii) [3 marks] Evaluate the double integral

$$\iint_R 3x^2 \sin(y) \, dA,$$

where R is the rectangle $R = \{(x, y) \mid 0 \leq x \leq 2 \text{ and } 0 \leq y \leq \pi\}$.

- (iii) [4 marks] By reversing the order of integration, evaluate the integral

$$\int_0^1 \int_x^1 (2 + 6x) \cos(y^2 + y^3) \, dy \, dx.$$

- (iv) (*beyond 2026 midterm scope*) [5 marks] Let E be the solid lying below the plane $z = 2 - x$ and above the triangular region in the xy -plane with vertices $(0, 0, 0)$, $(4, 0, 0)$, and $(0, 4, 0)$. Evaluate the triple integral

$$\iiint_E x \, dV.$$

Solution**Method 1: Direct calculation**

- (i) e^t increasing: extremise $g = xy$ on $h = 4x^2 + y^2 = 8$; ellipse compact (EVT), $\nabla h \neq \mathbf{0}$ on it, so Lagrange applies.

$$y = 8\lambda x, \quad x = 2\lambda y \implies x, y \neq 0 \quad (x = 0 \iff y = 0), \quad \lambda = \pm \frac{1}{4}, \quad y = \pm 2x \quad (\nabla g = \lambda \nabla h)$$

$$8x^2 = 8 \implies T = e^{2xy} = e^4 \text{ at } \pm(1, 2), \quad e^{-4} \text{ at } \pm(1, -2) \quad (h = 8)$$

$$\boxed{T_{\max} = e^4 \text{ at } (1, 2), (-1, -2); \quad T_{\min} = e^{-4} \text{ at } (1, -2), (-1, 2).}$$

- (ii) $R = [0, 2] \times [0, \pi]$, Fubini:

$$\iint_R 3x^2 \sin y \, dA = \int_0^2 3x^2 \, dx \int_0^\pi \sin y \, dy = [x^3]_0^2 [-\cos y]_0^\pi = \boxed{16}$$

- (iii) Region $0 \leq x \leq y \leq 1$, Fubini:

$$\begin{aligned} \int_0^1 \int_x^1 (2 + 6x) \cos(y^2 + y^3) \, dy \, dx &= \int_0^1 \int_0^y (2 + 6x) \cos(y^2 + y^3) \, dx \, dy \\ &= \int_0^1 (2y + 3y^2) \cos(y^2 + y^3) \, dy \\ &= \boxed{[\sin(y^2 + y^3)]_0^1 = \sin 2} \quad (d(y^2 + y^3)) \end{aligned}$$

- (iv) $0 \leq z \leq 2 - x$ needs $x \leq 2$: $E = \{0 \leq x \leq 2, 0 \leq y \leq 4 - x, 0 \leq z \leq 2 - x\}$, Fubini:

$$\iiint_E x \, dV = \int_0^2 \int_0^{4-x} \int_0^{2-x} x \, dz \, dy \, dx = \int_0^2 x(2-x)(4-x) \, dx = \boxed{4}$$

Method 2: AM–GM**(i)**

$$\begin{aligned}
 |xy| &= \frac{1}{2} |2x| |y| \leq \frac{1}{4} ((2x)^2 + y^2) = \frac{h}{4} = 2 & (\text{AM–GM}) \\
 \text{equality} \iff y = \pm 2x &\implies x = \pm 1 & (h = 8) \\
 &\implies xy = 2 \text{ at } \pm(1, 2), \quad xy = -2 \text{ at } \pm(1, -2) \\
 &\implies T = e^{2xy} \in [e^{-4}, e^4], \text{ attained there}
 \end{aligned}$$

UNOFFICIAL QRS marking scheme (17 marks)

- (i) **(5 marks)** 1: correct objective and constraint; 2: Lagrange equations and solutions; 1: classifies the signs; 1: states both extrema and all points.
- (ii) **(3 marks)** 1: correct iterated integral; 1: evaluates both factors; 1: obtains 16.
- (iii) **(4 marks)** 2: correct reversed bounds; 1: recognises the derivative; 1: obtains $\sin 2$.
- (iv) **(5 marks)** 1: identifies the nonnegative-height footprint; 2: correct height integral; 1: evaluates it; 1: obtains 4.

Question 2*(beyond 2026 midterm scope)***(17 Marks)****QUESTION 2.**

- (i) [4 marks] Use cylindrical coordinates to evaluate the triple integral

$$\iiint_E 2z \, dV,$$

where E is the solid above the hemisphere $z = \sqrt{4 - x^2 - y^2}$ and below the ellipsoid $z = \sqrt{8 - 2x^2 - 2y^2}$.

- (ii) [6 marks] The solid
- E
- is a spherical wedge such that

$$\iiint_E f(x, y, z) \, dV = \int_0^1 \int_x^{\sqrt{2-x^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{4-x^2-y^2}} f(x, y, z) \, dz \, dy \, dx.$$

The solid E can be described in spherical coordinates by $T_1 \leq \theta \leq T_2$, $P_1 \leq \phi \leq P_2$, and $R_1 \leq \rho \leq R_2$. Find the values of T_1, T_2, P_1, P_2, R_1 , and R_2 .

- (iii) Consider the transformation
- $u = x + y$
- and
- $v = y^2$
- .

- (a) [2 marks] Calculate the Jacobian
- $\frac{\partial(u, v)}{\partial(x, y)}$
- .

- (b) Using the transformation, one can evaluate the double integral

$$\iint_R 2y \cos((x + y)^3 - 3(x + y)) \, dA,$$

where R is the region bounded by the curves $x = 0$, $y = 1$, and $x + y = 2$. One finds that

$$\iint_R 2y \cos((x + y)^3 - 3(x + y)) \, dA = \int_a^b \int_c^d F(u, v) \, dv \, du.$$

- (I) [2 marks] Find the values of a, b, c, d . [Hint: they are not all constants].
 (II) [1 mark] Find the integrand $F(u, v)$.
 (III) [2 marks] It turns out that

$$\iint_R 2y \cos((x + y)^3 - 3(x + y)) \, dA = C_1 \sin(C_2)$$

for some constants C_1 and C_2 . Find the values of C_1 and C_2 .

Solution**Method 1: Direct calculation**

- (i)
- E
- :
- $0 \leq \theta \leq 2\pi$
- ,
- $0 \leq r \leq 2$
- (surfaces meet at
- $r = 2$
-),
- $\sqrt{4 - r^2} \leq z \leq \sqrt{8 - 2r^2}$
- :

$$\begin{aligned} \iiint_E 2z \, dV &= \int_0^{2\pi} \int_0^2 \int_{\sqrt{4-r^2}}^{\sqrt{8-2r^2}} 2zr \, dz \, dr \, d\theta \\ &= 2\pi \int_0^2 r(4 - r^2) \, dr = \boxed{8\pi} \quad ([z^2] = 4 - r^2) \end{aligned}$$

(ii)

$$\begin{aligned}
 x \geq 0, y \geq x &\implies \frac{\pi}{4} \leq \theta \leq \frac{\pi}{2} \\
 z \geq r &\implies 0 \leq \phi \leq \frac{\pi}{4} \quad (\text{cone } z = r) \\
 z \leq \sqrt{4 - r^2} &\implies 0 \leq \rho \leq 2 \quad (r \leq \sqrt{2}, x \leq 1 \text{ then automatic})
 \end{aligned}$$

$$T_1 = \frac{\pi}{4}, \quad T_2 = \frac{\pi}{2}, \quad P_1 = 0, \quad P_2 = \frac{\pi}{4}, \quad R_1 = 0, \quad R_2 = 2.$$

$$(iii) \quad (a) \quad \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} 1 & 1 \\ 0 & 2y \end{vmatrix} = \boxed{2y}$$

(b)(I) R : $0 \leq x \leq 2 - y$, $1 \leq y \leq 2$; $y \geq 1 > 0$, so $(x, y) \mapsto (u, v)$ is one-to-one.

$$y = 1, \quad x = 0, \quad x + y = 2 \mapsto v = 1, \quad v = u^2, \quad u = 2$$

$$\boxed{a = 1, \quad b = 2, \quad c = 1, \quad d = u^2.}$$

$$(b)(II) \quad dA = du \, dv / |2y| \text{ (change of variables), so } 2y \text{ cancels: } \boxed{F(u, v) = \cos(u^3 - 3u)}$$

(b)(III)

$$\begin{aligned}
 \int_1^2 \int_1^{u^2} \cos(u^3 - 3u) \, dv \, du &= \int_1^2 (u^2 - 1) \cos(u^3 - 3u) \, du = \frac{1}{3} [\sin(u^3 - 3u)]_1^2 \\
 &= \frac{1}{3} (\sin 2 - \sin(-2)) = \frac{2}{3} \sin 2 \implies \boxed{C_1 = \frac{2}{3}, \quad C_2 = 2}
 \end{aligned}$$

UNOFFICIAL QRS marking scheme (17 marks)

- (i) **(4 marks)** 1: cylindrical bounds; 1: Jacobian and integrand; 1: reduces the integral; 1: obtains 8π .
- (ii) **(6 marks)** 2: azimuthal bounds; 2: polar-angle bounds; 1: radial bounds; 1: all constants correct.
- (iii) **(7 marks)**
- (a) **(2 marks)** 1: forms determinant; 1: obtains $2y$.
- (b) **(5 marks)**
- (I) **(2 marks)** 1: maps the three boundaries; 1: gives all bounds.
- (II) **(1 mark)** correct transformed integrand.
- (III) **(2 marks)** 1: evaluates the one-variable integral; 1: gives C_1, C_2 .

Question 3**(16 Marks)***(beyond 2026 midterm scope)***QUESTION 3.**

- (i) [4 marks] Let
- C
- be the part of the curve
- $y = x^2$
- oriented from
- $(1, 1)$
- to
- $(2, 4)$
- . Evaluate

$$\int_C xy \, dx + (x + y) \, dy.$$

- (ii) [5 marks] Evaluate the line integral of
- $f(x, y, z) = x + y + z$
- over the straight-line segment from
- $(3, 1, 2)$
- to
- $(2, -3, -1)$
- .

- (iii) Consider the vector field
- $\mathbf{F} = \langle 4z^2, \cos z, 8xz - y \sin z + 1 \rangle$
- .

- (a) [4 marks] Show that
- $\mathbf{F}$
- is conservative by finding a potential function
- f
- for
- $\mathbf{F}$
- .

- (b) [3 marks] Evaluate

$$\int_C \mathbf{F} \cdot d\mathbf{r},$$

where C is the helix $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ for $0 \leq t \leq \pi/2$.**Solution****Method 1: Direct calculation**

- (i)
- C
- :
- $\mathbf{r}(x) = \langle x, x^2 \rangle$
- ,
- $1 \leq x \leq 2$
- ,
- $dy = 2x \, dx$
- :

$$\begin{aligned} \int_C xy \, dx + (x + y) \, dy &= \int_1^2 (x^3 + (x + x^2)2x) \, dx \\ &= \int_1^2 (3x^3 + 2x^2) \, dx = \boxed{\frac{191}{12}} \end{aligned}$$

- (ii)
- $\mathbf{r}(t) = \langle 3, 1, 2 \rangle + t\langle -1, -4, -3 \rangle$
- ,
- $0 \leq t \leq 1$
- ;
- $f(\mathbf{r}(t)) = 6 - 8t$
- ,
- $\|\mathbf{r}'\| = \sqrt{26}$
- :

$$\int_C f \, ds = \sqrt{26} \int_0^1 (6 - 8t) \, dt = \boxed{2\sqrt{26}}$$

- (iii) (a)

$$\begin{aligned} f_x = 4z^2 &\implies f = 4xz^2 + g(y, z) \\ f_y = \cos z &\implies g = y \cos z + h(z) \\ f_z = 8xz - y \sin z + h'(z) &= 8xz - y \sin z + 1 \implies h' = 1 \end{aligned}$$

$$\nabla f = \mathbf{F} \text{ (conservative)} : \quad \boxed{f(x, y, z) = 4xz^2 + y \cos z + z}$$

- (b)
- C
- :
- $(1, 0, 0) \rightarrow (0, 1, \pi/2)$
- ; fundamental theorem for line integrals:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = f(0, 1, \frac{\pi}{2}) - f(1, 0, 0) = \frac{\pi}{2} - 0 = \boxed{\frac{\pi}{2}}$$

UNOFFICIAL QRS marking scheme (16 marks)

- (i) **(4 marks)** 1: parametrises/orients the curve; 1: obtains dy ; 1: forms the integrand; 1: obtains $191/12$.
- (ii) **(5 marks)** 1: parametrises the segment; 1: finds ds ; 2: substitutes into the scalar line integral; 1: obtains $2\sqrt{26}$.
- (iii) **(7 marks)**
 - (a) **(4 marks)** 1: integrates the first component; 1: matches the second; 1: matches the third; 1: valid potential.
 - (b) **(3 marks)** 1: identifies endpoints; 1: applies the theorem; 1: obtains $\pi/2$.